**Secure By Design**

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COURSE 1.10 / COMPANION WORKBOOK

# Collect the record on SBD-SYSLOG01

## Design the logging pipeline and build its collector

The first of three logging courses: design the whole collect → protect → analyze pipeline, then build its first stage — advanced audit policy on the DCs and admin hosts, source-initiated Windows Event Forwarding from every existing Windows system and a syslog intake on SBD-SYSLOG01, health checks with a named owner, and proof that real events arrive with the right names and times.

MATERIALS EDITION 2.0 / COURSE 1.10

Updated 14 October 2026 · sbd-logging-pipeline

NAME _____ / DATE _____

START HERE / YOUR LEARNING PATH

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Read each lesson alongside the interactive guide and the video. Use the record tables to write down what you actually observe — the workbook is your evidence, not the guide.


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04 Keep the collector trustworthy

05 Prove collection and hand off

Course review and sign-off


> **HOW TO USE IT** Each lesson page has the same shape: what you will learn, the build steps, the commands that verify them, and a record table. Fill the table from your own environment — never from the sample values.

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# Before you start

This is the first of three logging courses. Course 1.10 designs the whole collect → protect → analyze pipeline and builds the collect stage on SBD-SYSLOG01; course 1.11 protects the record on SBD-AUDIT01, and course 1.12 analyzes it in SBD-SPLUNK01. Collection is configuration you can show: a subscription, a listener, and events with the right source names and times.

## Prerequisites

- ☐ Complete courses 1.1–1.9: every Windows system they built — both DCs, SBD-SUBCA01, SBD-PKI-MGMT01, SBD-DHCP01, and the three admin seats — becomes a source; the tiered admin plane (course 1.8) means the collector is administered only from VLAN 60 (SBD-ADMIN-SRV01, Tier 1); and SBD-REPO01 (10.0.70.20, course 1.9) is running and becomes this course's first Linux source.
- ☐ Provision SBD-SYSLOG01 (10.0.20.20) as Windows Server joined to sbd.test in the Servers OU, so course 1.5's member-server STIG policies apply from first boot, with a separate data volume (D: in this sample) for the logs. Turn off the hypervisor's time synchronization and check that w32tm /query /source names a domain controller (course 1.3).
- ☐ Have the network's service-path register, the course 1.6 SBD Server Authentication template, and an approved syslog receiver for Windows (staged across the air gap and hashed, as in course 1.4) ready: Windows has no built-in syslog listener.

## What this course builds

- Design all three logging stages: retention, readers, streams, and register rows.
- Write the designed audit events on the DCs and admin hosts through GPO.
- Forward every existing Windows system's events by source-initiated subscription and take syslog from the router and SBD-REPO01.
- Keep the collector healthy and prove collection with events you cause.

## Systems in scope

| RECORD | YOUR OBSERVATION |
|---|---|
| SBD-SYSLOG01 — name, address, OS build | |
| SBD-DC01 — name, address, OS build | |
| SBD-DC02 — name, address, OS build | |
| SBD-SUBCA01 — name, address, OS build | |
| SBD-PKI-MGMT01 — name, address, OS build | |

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| RECORD | YOUR OBSERVATION |
| --- | --- |
| SBD-DHCP01 — name, address, OS build | |
| SBD-REPO01 — name, address, OS build | |

LESSON 01 / ABOUT 25 MIN

# Design collect, protect, analyze

> **WHY THIS MATTERS** Three systems, three distinct jobs: SBD-SYSLOG01 gathers, SBD-AUDIT01 preserves, SBD-SPLUNK01 answers questions. This course builds the first; courses 1.11 and 1.12 build the other two — so decide now, for all three, what is collected, how long it lives, who may touch it, and where each stream is picked up.

## In this lesson you will

- Assign the three roles, the course that builds each, and the first collection scope.
- Decide retention, readers, and the no-delete rule per stage.
- Name the streams and register every pipeline path.

## Follow the build

1. Write the role sentence for each host and the course that builds it: SBD-SYSLOG01 (course 1.10) receives Windows events and syslog; SBD-AUDIT01 (course 1.11) holds the protected, long-retention copy; SBD-SPLUNK01 (course 1.12) indexes a working copy for search and alerting. One copy to keep, one copy to question.
2. Scope the first collection to what already produces evidence: the Security log and the Windows Time events course 1.3 keeps from every Windows system that exists — both DCs, SBD-SUBCA01, SBD-PKI-MGMT01, SBD-DHCP01, and the three admin seats — plus the core router's deny log and SBD-REPO01's syslog. The fleet joins in course 1.13, and every later server in the course that builds it.
3. Decide the AU answers per stage (sample values; your records requirement decides): the collector keeps at least 14 days locally as a buffer, administered by Tier 1 from SBD-ADMIN-SRV01; the repository keeps one year, written only by a write-only archival account, read by a named auditors group, and deleted by nobody outside the custodian's logged retention process, with independent approval (rule A04); analysis keeps 90 days searchable, read by analysts.
4. Name the two streams and where they are picked up: Windows events land in the ForwardedEvents log (file on D:\Logs\ForwardedEvents) and syslog in daily files per sender under D:\Logs\Syslog. Course 1.11's archival job and course 1.12's forwarder both read from these two places; nothing else writes there.
5. Add the register rows with owners: Windows sources to the collector (F17, WinRM 5985, source-initiated), Linux syslog over TLS (F07, TCP 6514), the core router's deny log (UDP 514, which the router already sends to the collector's address as part of the network's configuration — this course only receives it), the archive path (F11, SMB 3 with encryption) and the forwarder path (F10, TCP 9997 with TLS). The last two stay unused until courses 1.11 and 1.12.

## Check your result

```
# From AdminJumpBox01 (Tier 0): what the DCs audit today.
Invoke-Command SBD-DC01, SBD-DC02 { auditpol /get /category:* /r } | Out-File C:\Evidence\
logging-pipeline\DCs-auditpol-before.csv
# From SBD-ADMIN-SRV01 (Tier 1): no collector runs yet.
Invoke-Command SBD-SYSLOG01 { Get-Service Wecsvc | Select-Object Status, StartType }
```

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**WHAT SUCCESS LOOKS LIKE** The design names each stage's job, building course, scope, retention, and readers; the no-delete rule is written; the two streams and their pickup folders are named; and the register carries the five pipeline paths with owners.

## Your record

Evidence to keep: Keep the pipeline design sheet, the retention and access table, the stream and pickup definitions, and the new register rows.

| RECORD | YOUR OBSERVATION |
|---|---|
| Host(s) and time collected | |
| Observed result | |
| Differences from the plan | |
| Evidence file names | |
| Open items or findings | |

## Think it through

Why separate the repository from the analysis system?

**REMEMBER** Collector gathers (1.10), repository preserves (1.11), analysis answers (1.12). Retention, readers, and the no-delete rule are written for every stage. Two named streams, their pickup folders, and five registered paths.

Microsoft · Use Windows Event Forwarding to help with intrusion detection

NIST · SP 800-53 Rev. 5 security controls (AU family)

LESSON 02 / ABOUT 25 MIN

# Turn on the right auditing

> **WHY THIS MATTERS** Forwarding garbage faster is not monitoring. Set the advanced audit policy so the events that tell the network's story — sign-ins, account changes, directory changes, policy changes — actually get written, at or above the STIG values course 1.4 measured.

## In this lesson you will

- Apply the advanced audit policy subcategories the design needs.
- Reconcile audit settings with the course 1.5 STIG Group Policy objects.
- Verify effective auditing on the DCs and admin hosts.

## Follow the build

1. Review what the Defense Information Systems Agency (DISA) Group Policy objects (GPOs) linked in course 1.5 already audit — most of the advanced audit policy arrived with the STIGs, including the setting that makes subcategories override the old category policy. This lesson reconciles and fills gaps rather than starting over.
2. Confirm the subcategories the design depends on are at least at the STIG values on the DCs: Credential Validation, Logon, Special Logon, User Account Management, Security Group Management, Directory Service Changes, Audit Policy Change, and Security State Change, which writes event 4616 for every clock change course 1.3 records — success and failure where the design says so. Directory Service Changes writes event 5136 only for objects whose auditing entries ask for it, so check the domain's auditing entries too.
3. Extend the same policy to the Admin OU's computer sub-OUs from course 1.8, so privileged sessions on the admin hosts write their sign-in trail, and confirm that SBD-SUBCA01, SBD-PKI-MGMT01, and SBD-DHCP01 already carry the member-server STIG values.
4. Make every addition a separate, documented GPO linked above the STIG GPO, never an auditpol /set on one server. Advanced audit settings merge across GPOs per subcategory, and only a conflicting subcategory is decided by precedence — record each addition in the course 1.5 exception record, kept on AdminJumpBox01.

## Check your result

```
# From AdminJumpBox01 (Tier 0): effective audit policy and the policy result on a DC.
Invoke-Command SBD-DC01 { auditpol /get /category:* /r } | Out-File C:\Evidence\logging-pipeline\SBD-DC01-auditpol.csv
Invoke-Command SBD-DC01 { auditpol /get /subcategory:"Directory Service Changes,Security State Change" }
gpresult /s SBD-DC01 /scope computer /h C:\Evidence\logging-pipeline\SBD-DC01-gpresult.html
```

> **WHAT SUCCESS LOOKS LIKE** Effective audit policy on the DCs and admin hosts covers the designed subcategories at or above the STIG values, applied through GPO, with every addition documented.

## Your record

Evidence to keep: Keep the auditpol output per host class, the GPO report, and the documented additions.

| RECORD | YOUR OBSERVATION |
| --- | --- |
| Host(s) and time collected | |
| Observed result | |
| Differences from the plan | |
| Evidence file names | |
| Open items or findings | |

## Think it through

Why start from the course 1.5 STIG GPOs instead of a fresh audit policy?

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> **REMEMBER** Audit policy is reconciled with the STIG baseline, not reinvented. DCs and admin hosts write the designed event families. Additions are GPO-visible and documented.

Microsoft · Use Windows Event Forwarding to help with intrusion detection

Microsoft · Auditpol

Microsoft · Advanced security auditing FAQ

Microsoft · Advanced audit policy configuration

LESSON 03 / ABOUT 35 MIN

# Build the collector

> **WHY THIS MATTERS** SBD-SYSLOG01 becomes the funnel: Windows Event Forwarding brings in the events of every Windows system that exists by source-initiated subscription, and a syslog receiver takes the core router's denies and, from lesson 5, the Linux servers over TLS — one host where the network's record first lands.

## In this lesson you will

- Configure the Windows Event Collector service and a source-initiated subscription.
- Enroll the sources by GPO with read access to their Security logs.
- Stand up the syslog intake for UDP 514 and TLS on TCP 6514.

## Follow the build

1. On SBD-SYSLOG01, run wecutil qc /q, move the ForwardedEvents log file to D:\Logs\ForwardedEvents, and create the subscription SBD-ForwardedSecurity from an XML file: SourceInitiated, MinLatency delivery (push, 30-second batches), into ForwardedEvents, with two queries — the whole Security log (event 4616 included), and the Microsoft-Windows-Time-Service events 12, 35, 36, 37, 129, and 142 from the System log that course 1.3 keeps. Allow one computer group, SBD Event Sources, created from AdminJumpBox01 and holding every Windows system that exists now: both DCs, SBD-SUBCA01, SBD-PKI-MGMT01, SBD-DHCP01, and the three admin seats. The default allows Domain Computers, which does not include domain controllers.
2. From AdminJumpBox01, enroll the sources with one GPO linked to the Domain Controllers, Tier 0 Servers, Servers, and Admin sub-OUs and security-filtered to SBD Event Sources (Authenticated Users keep read), so the collector and later servers stay out until a course adds them: the subscription manager Server=http://SBD-SYSLOG01.sbd.test:5985/wsman/SubscriptionManager/WEC,Refresh=60, the WinRM service set to automatic, and NETWORK SERVICE in Event Log Readers — on the DCs through the domain's built-in group, on the other sources as a local group preference. Restart each source so the group changes apply; the sources push, the collector reads nothing.
3. Allow WinRM 5985 inbound on the collector only from the enrolled sources: VLANs 10 and 60, SBD-SUBCA01 and SBD-PKI-MGMT01 (10.0.30.20 and 10.0.30.30), and SBD-DHCP01 (10.0.20.10), which shares VLAN 20, so only this host firewall scopes it. Rule F17 covers every Windows member; each later course widens this scope when it enrolls new sources. Kerberos authenticates both ends and encrypts the forwarded events, even over HTTP.
4. Install the approved syslog receiver as a service: UDP 514 for the core router, which already sends its deny log to this address, and TCP 6514 with TLS using an SBD Server Authentication certificate for SBD-SYSLOG01.sbd.test, requested from the course 1.6 issuing CA through its process after adding SBD-SYSLOG01 to the SBD TLS Servers group. Write daily files per sender under D:\Logs\Syslog, and allow each port only from its registered senders.
5. Confirm the register rows are the only paths in use: WinRM from the enrolled sources, syslog from the core, and nothing listening that the design didn't ask for.

## Check your result

```
# On SBD-SYSLOG01 (Tier 1 session from SBD-ADMIN-SRV01): the subscription, the first
events, and the listeners.
```

```
wecutil es
wecutil gs SBD-ForwardedSecurity
Get-WinEvent -LogName ForwardedEvents -MaxEvents 20 | Select-Object TimeCreated, Id,
MachineName
Get-NetUDPEndpoint -LocalPort 514
Get-NetTCPConnection -LocalPort 6514,5985 -State Listen
# On each source, from its tier's seat: Network Service is in Event Log Readers, and the
forwarder connected (events 104 and 100).
net localgroup "Event Log Readers"
Get-WinEvent -LogName 'Microsoft-Windows-Forwarding/Operational' -MaxEvents 5 | Select-
Object TimeCreated, Id, Message
```

> **WHAT SUCCESS LOOKS LIKE** The subscription is enabled, events from every member of SBD Event Sources — Security and Time-Service — appear in ForwardedEvents with correct source machine names, router deny entries arrive in the syslog files, and only the registered ports listen.

## Your record

Evidence to keep: Record the subscription XML and wecutil gs output, the enrolling GPO report, the first forwarded event per source, a received router deny line, and the listener list.

| RECORD | YOUR OBSERVATION |
|---|---|
| Host(s) and time collected | |
| Observed result | |
| Differences from the plan | |
| Evidence file names | |
| Open items or findings | |

## Think it through

Why source-initiated subscriptions rather than the collector pulling?

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> **REMEMBER** Sources push by subscription; the collector reaches into nothing. Windows sources and router records land in two named streams. Only registered paths carry the pipeline.

Microsoft · Use Windows Event Forwarding to help with intrusion detection

Microsoft · Setting up a source initiated subscription

Microsoft · Wecutil (Windows Event Collector)

RFC 5425 · Transport Layer Security (TLS) Transport Mapping for Syslog

LESSON 04 / ABOUT 25 MIN

# Keep the collector trustworthy

> **WHY THIS MATTERS** A collector that silently stops is worse than none, because everyone assumes it works. Watch the subscription's runtime status, the sources that went quiet, the log's size and overwrite behavior, the disk, and the syslog intake — and decide who is told when collection fails.

## In this lesson you will

- Read subscription runtime status and find silent sources.
- Size the ForwardedEvents log and data volume for the design's buffer.
- Define what a collection failure is and who responds.

## Follow the build

1. Read wecutil gr SBD-ForwardedSecurity: each source shows a runtime status and its last heartbeat. A source that is inactive, or whose last heartbeat is older than the subscription's heartbeat interval (wecutil gs shows it), has gone silent — check its Microsoft-Windows-Forwarding/Operational log first.
2. Set the ForwardedEvents log's size and behavior with wevtutil sl: overwrite the oldest events (/rt:false) so new events are never refused, sized so the oldest event is always older than the 14-day buffer. Overwriting is safe only while courses 1.11 and 1.12 pick events up well inside that window; a full log in retain mode would drop new events instead.
3. Check the data volume and the syslog intake: free space on D:, both listeners up, and today's file growing for every registered sender. A sender with no file today is a silent source too.
4. Write the failure response (AU-5): a stopped Windows Event Collector service, an inactive source, the oldest event younger than the buffer, a volume over 80 percent full, or a silent syslog sender. The logging owner is told the same day, restores collection, and records the gap window. Until course 1.12 turns these into alerts, run the check daily and keep each result.

## Check your result

```
wecutil gr SBD-ForwardedSecurity
wevtutil sl ForwardedEvents /ms:4294967296 /rt:false
wevtutil gl ForwardedEvents
Get-WinEvent -LogName ForwardedEvents -Oldest -MaxEvents 1 | Select-Object TimeCreated
Get-Volume -DriveLetter D | Select-Object SizeRemaining, Size
Get-ChildItem D:\Logs\Syslog -Recurse -File | Sort-Object LastWriteTime -Descending |
Select-Object -First 5 FullName, LastWriteTime
```

> **WHAT SUCCESS LOOKS LIKE** Every enrolled source is active with a recent heartbeat, ForwardedEvents overwrites oldest-first and holds more than 14 days, the data volume has headroom, every registered syslog sender wrote today, and the failure response names its owner.

## Your record

Evidence to keep: Keep the runtime status output, the log configuration, the oldest-event time, the volume figures, and the written failure response with its first daily check.

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| RECORD | YOUR OBSERVATION |
| --- | --- |
| Host(s) and time collected | |
| Observed result | |
| Differences from the plan | |
| Evidence file names | |
| Open items or findings | |

## Think it through

A DC stopped forwarding last night. Where does it first show on the collector?

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> **REMEMBER** Runtime status and heartbeats show silent sources. The log is sized past the buffer and overwrites oldest-first. Collection failures have a definition, an owner, and a record.

Microsoft · Use Windows Event Forwarding to help with intrusion detection

Microsoft · Wecutil (Windows Event Collector)

Microsoft · Wevtutil

NIST · SP 800-53 Rev. 5 security controls (AU family)

LESSON 05 / ABOUT 30 MIN

# Prove collection and hand off

> **WHY THIS MATTERS** Prove the collector with events you cause on purpose: a failed sign-in, a denied probe, and a test message from SBD-REPO01, the first Linux source. Each must land with the right source name and time — then the cadence scans close the course.

## In this lesson you will

- Trace a failed sign-in and a router deny to the collector.
- Enroll SBD-REPO01 as the first syslog-over-TLS source.
- Baseline SBD-SYSLOG01, rescan the DCs, and write the handoff.

## Follow the build

1. Generate one failed sign-in on SBD-DC01 with a wrong password for a tier admin account. Find event 4625 in the DC's Security log, then in ForwardedEvents with MachineName SBD-DC01 and the same time; compare the delay with the 30-second delivery setting.
2. Probe one unregistered port across the core — from SBD-ADMIN-SRV01, TCP 23 on SBD-DHCP01 — and find the router's deny line in the syslog files with the right source, destination, port, and time. Attempt and deny now form an evidence pair.
3. Enroll SBD-REPO01 (rule F07): from its own approved package set, install rsyslog's TLS module (rsyslog-gnutls, or your distribution's equivalent), point its TLS settings at the system trust store, which holds the course 1.6 root that course 1.9 placed there, and forward to SBD-SYSLOG01.sbd.test on TCP 6514, checking the collector's certificate name. Add REPO01 to the 6514 sender list, send a test message with logger, and find it in REPO01's daily file.
4. Run the cadence: SBD-SYSLOG01's first SCAP baseline with the course 1.4 benchmark and tool versions, run locally in a Tier 1 session from SBD-ADMIN-SRV01 with SCC copied from that seat's course 1.8 copy over PowerShell remoting and its hash checked, and a rescan of both DCs from AdminJumpBox01, attributing the deltas to lesson 2's audit policy.
5. Write the handoff to course 1.11: the two streams, their pickup folders, the 14-day buffer the archive job must stay inside, the gap windows the daily check records (those days need re-exporting once a source replays its backlog), and the F11 row waiting for SBD-AUDIT01.

## Check your result

```
# On SBD-SYSLOG01 (Tier 1 session): the failed sign-in, and the router's deny of the probe
to SBD-DHCP01.
Get-WinEvent -FilterHashtable @{LogName='ForwardedEvents'; Id=4625; StartTime=(Get-
Date).AddHours(-1)} | Select-Object TimeCreated, MachineName
Select-String -Path D:\Logs\Syslog\*\*.log -Pattern '10\.0\.20\.10' | Select-Object -Last
3
# On SBD-REPO01: check the configuration and the TLS session, then send the test message.
rsyslogd -N1
ss -tn 'dport = :6514'
logger -t sbd-collector-proof 'SBD-REPO01 syslog over TLS test'
# Back on SBD-SYSLOG01: the test message in REPO01's daily file.
Select-String -Path D:\Logs\Syslog\*\*.log -Pattern 'sbd-collector-proof' | Select-Object
-First 3
# On SBD-ADMIN-SRV01: this course's evidence, hashed.
Get-FileHash -Algorithm SHA256 C:\Evidence\logging-pipeline\*
```

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**WHAT SUCCESS LOOKS LIKE** The failed sign-in, the router deny, and REPO01's test message each appear on the collector with the right source and a time that matches the source; SBD-SYSLOG01 has its first baseline; and the DC rescan deltas are attributed to the audit policy.

## Your record

Evidence to keep: Keep the three traces with their timestamps, REPO01's forwarding configuration, the scan set with hashes, and the handoff note to course 1.11.

| RECORD | YOUR OBSERVATION |
|---|---|
| Host(s) and time collected | |
| Observed result | |
| Differences from the plan | |
| Evidence file names | |
| Open items or findings | |

## Think it through

What does a deliberately caused event prove that configuration checks don't?

**REMEMBER** A failed sign-in, a deny, and a Linux message all landed with the right names and times. SBD-REPO01 is the first syslog-over-TLS source. Baselines, rescans, and the handoff to course 1.11 close the course.

Microsoft · Wecutil (Windows Event Collector)

RFC 5425 · Transport Layer Security (TLS) Transport Mapping for Syslog

DISA · SRG/STIG tools (SCC and STIG Viewer)

NIST · SP 800-53 Rev. 5 security controls (AU family)

# Course review and sign-off

Before moving on, confirm each lesson left evidence behind — and that what remains open is written down, not remembered.

## Lesson completion

| LESSON | EVIDENCE FILED | DATE |
|---|---|---|
| 01 Design collect, protect, analyze | ☐ yes | |
| 02 Turn on the right auditing | ☐ yes | |
| 03 Build the collector | ☐ yes | |
| 04 Keep the collector trustworthy | ☐ yes | |
| 05 Prove collection and hand off | ☐ yes | |

## Open items carried forward

## Next in the learning path

That completes course 1.10. Course 1.11 gives the record its protected home on SBD-AUDIT01, picking up the two streams this course collects.

> **KEEP IT HONEST** Lesson completion records learning activity; it is not a system authorization decision. Your observations, not the sample values, establish what happened in your environment.

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